
Octoboe meeting

Road Grade-Aware Energy Consumption Modeling for EVs

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○○ Mechanical Engineering,
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October 2025

◆ List of researchers

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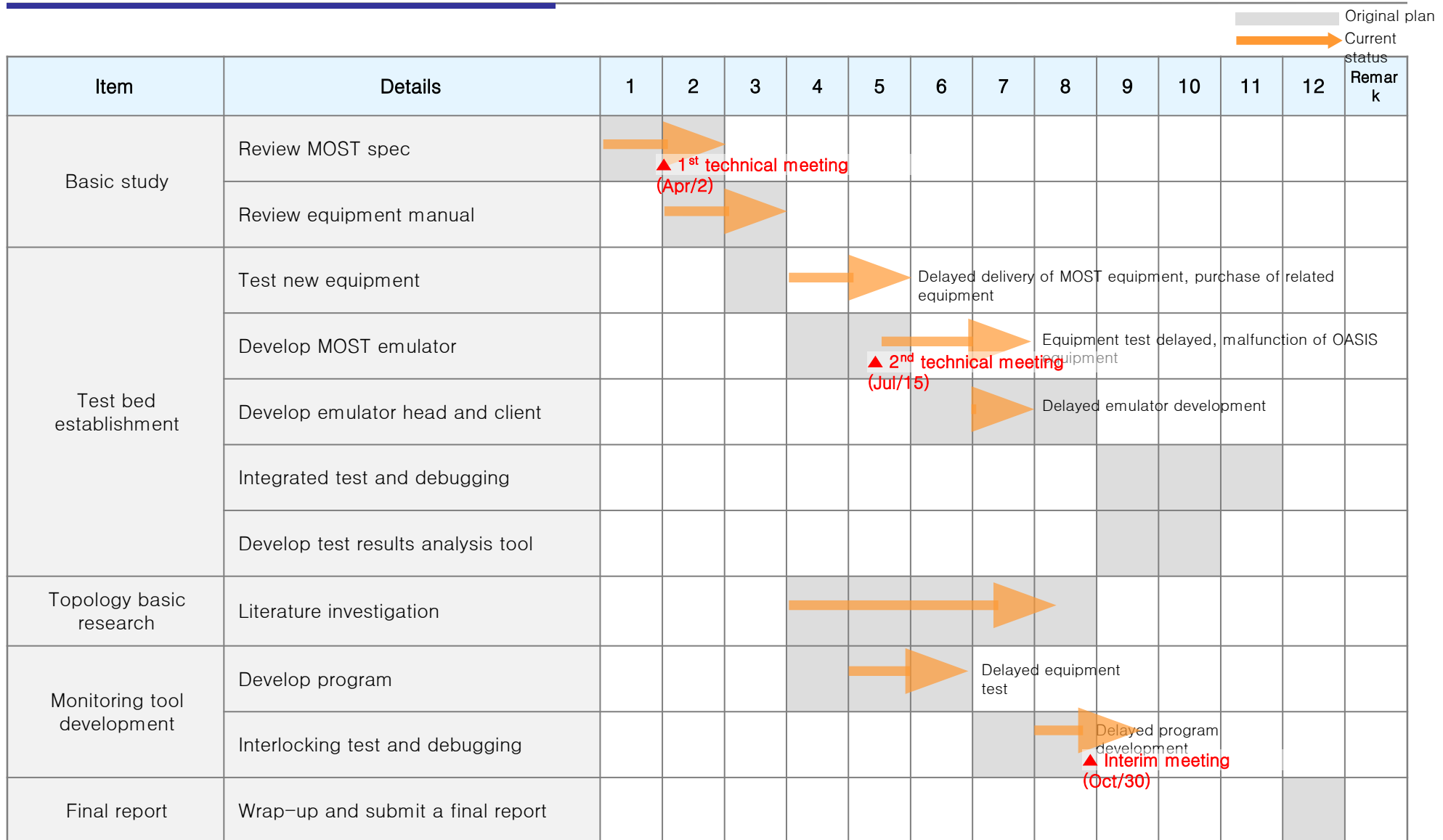
I. Background of the research

- Motivation:
 - EV range prediction is critical for adoption and user confidence (fight range anxiety).
 - Current models lack accuracy on hilly roads.
- Problem Statement
 - Road grade has a significant impact on energy consumption
 - Most models are generic, not vehicle-specific
 - Identify the main parameters affecting the energy consumption of EV on sloped roads
- Necessity
 - Develop and validate a physics/empirical model tailored for real vehicles on real roads
 - Incorporate an grade aware energy model in a car path planner

II. Research goal and expected effect

Research goal	• Collect real world data of energy consumption, road grade and other relevant parameters • Analyze and identify the main parameters that affect the energy consumption on roads with different grades • Build and validate a road grade-aware energy model for electric vehicles
Expected effect	• Improve the accuracy of trajectory planning with grade aware models • lower the probability of energy distress of an EV due to higher than expected road grades

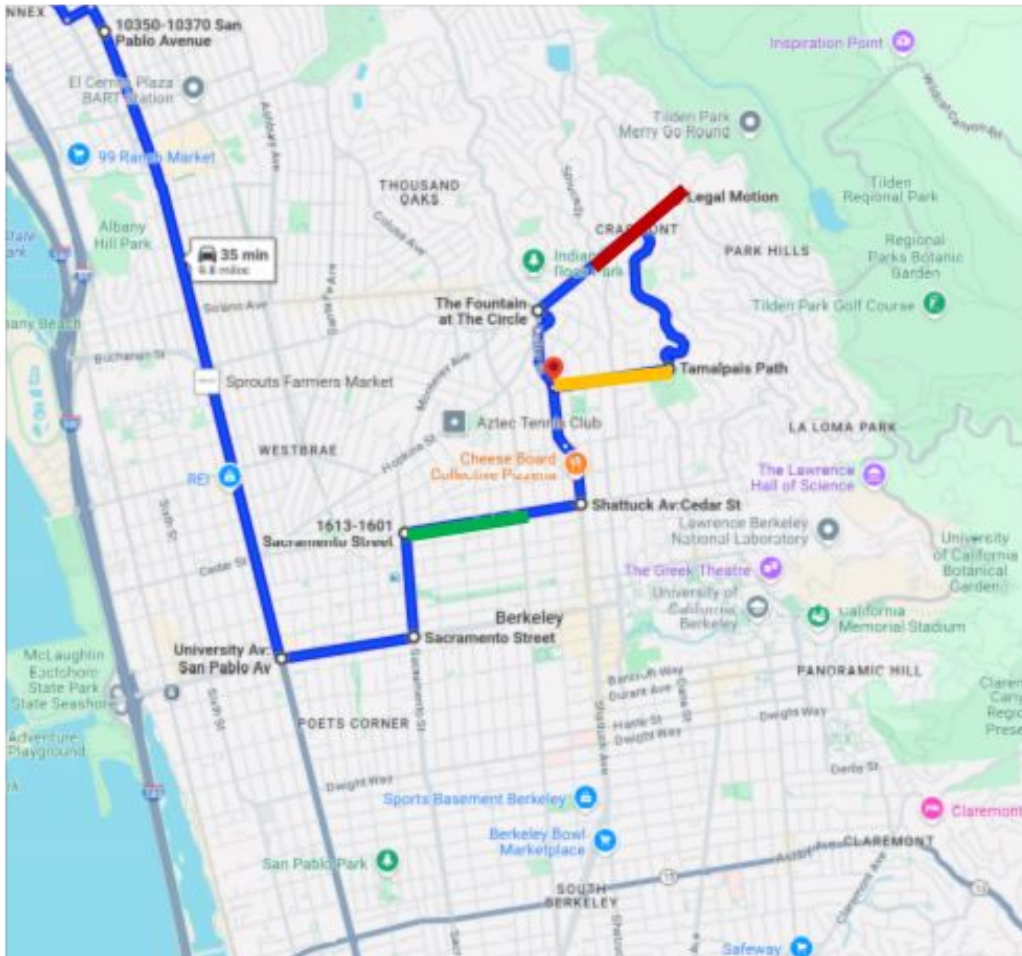
III. Research progress



IV. Research contents

Collected Data with full metry on a tailored drive in hilly Berkeley

Sample Data Collection



Flat Road

Moderate Road Grade

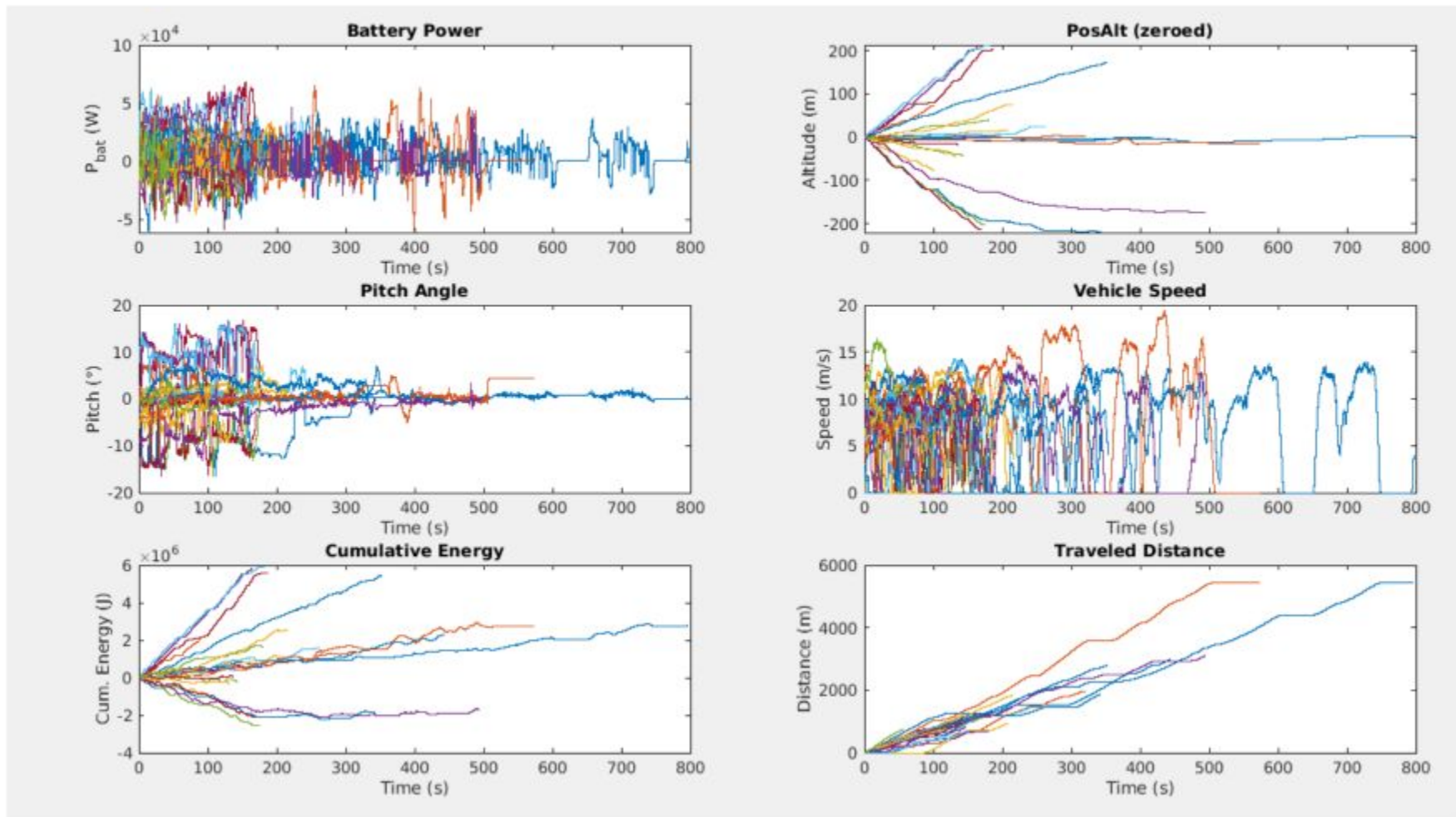
Steep Road Grade

- Total traveled distance: ~20 Km
- Normalize all Recordings to 1000m travel distance
- About 30 useful data points
- Normal Driving Mode
- Regen Level 1

IV. Research contents

Filter the data and normalize each valid data point to 1000m

Sample Data Collection

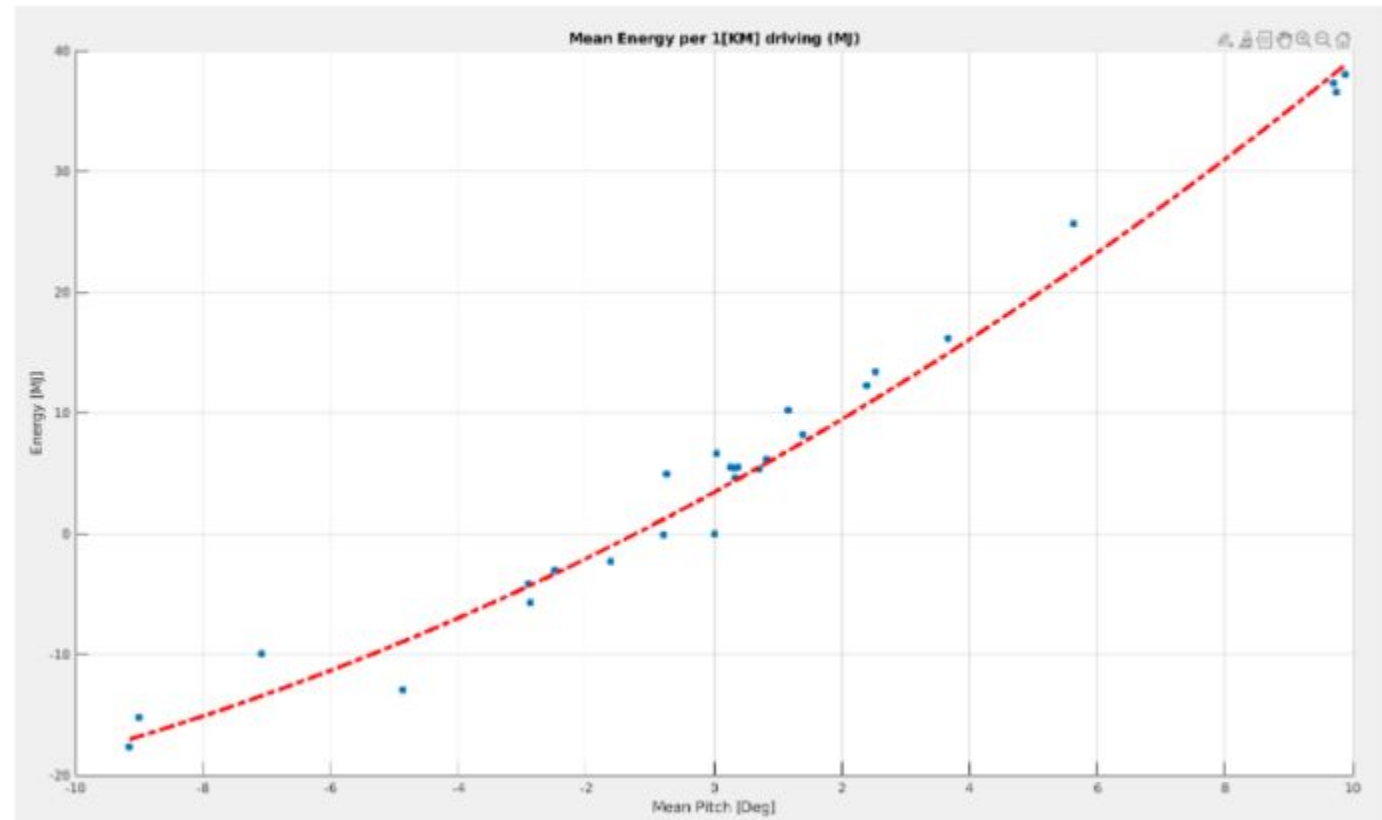


IV. Research contents

Identify the main parameters that affect energy consumption and suggest a prediction model

Data Analysis

- The mean Energy it takes to drive 1[Km] at a specific road slope
- The measurements seem to fit a parabolic model
- Going uphill and down hill is not the same (not bias)
-

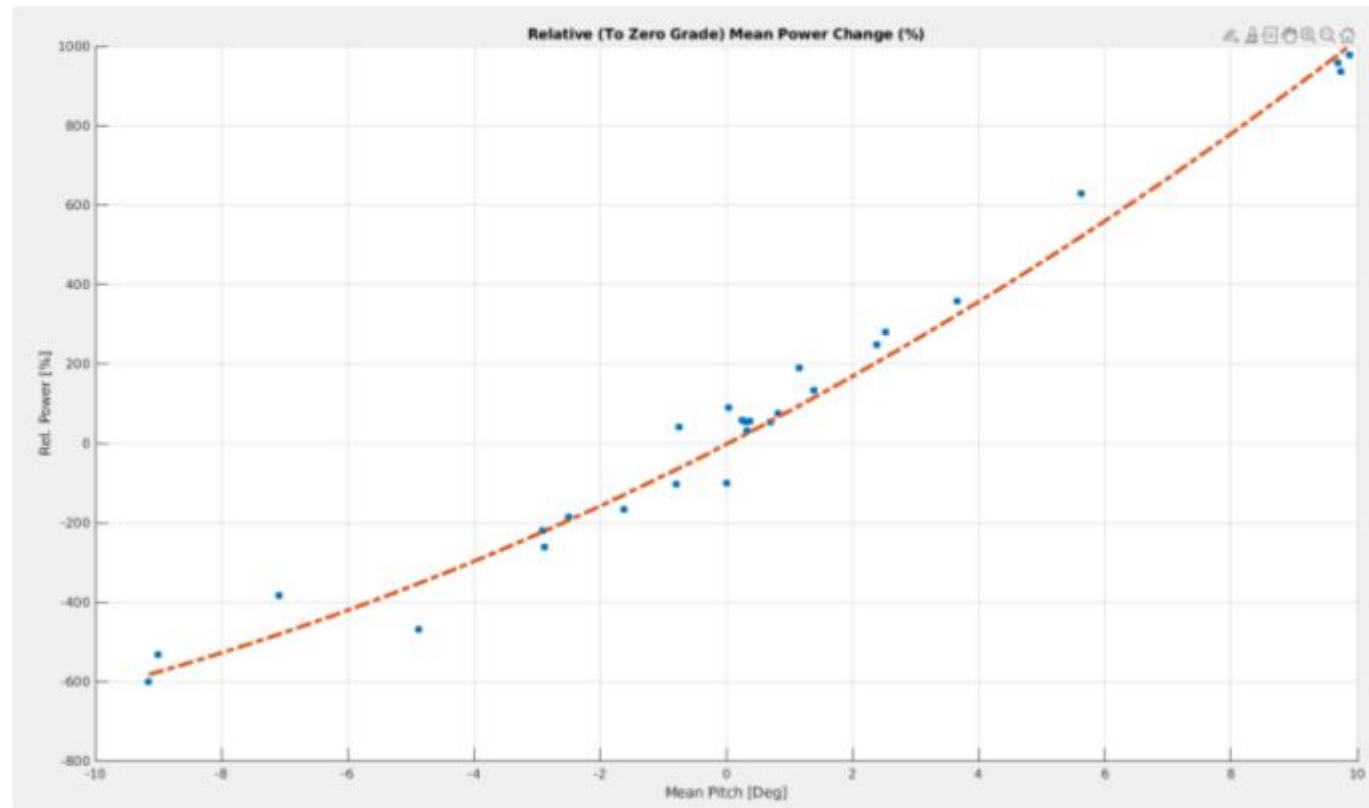


IV. Research contents

Data analysis – The effect of sloped roads on EV energy consumption is dramatic and should be taken into account, can increase power consumption by hundreds of percents

Data Analysis

- The relative change in power consumption
- Driving on a flat road is not the same as driving uphill and then downhill
- It is a dominant factor in planning for EV route planning



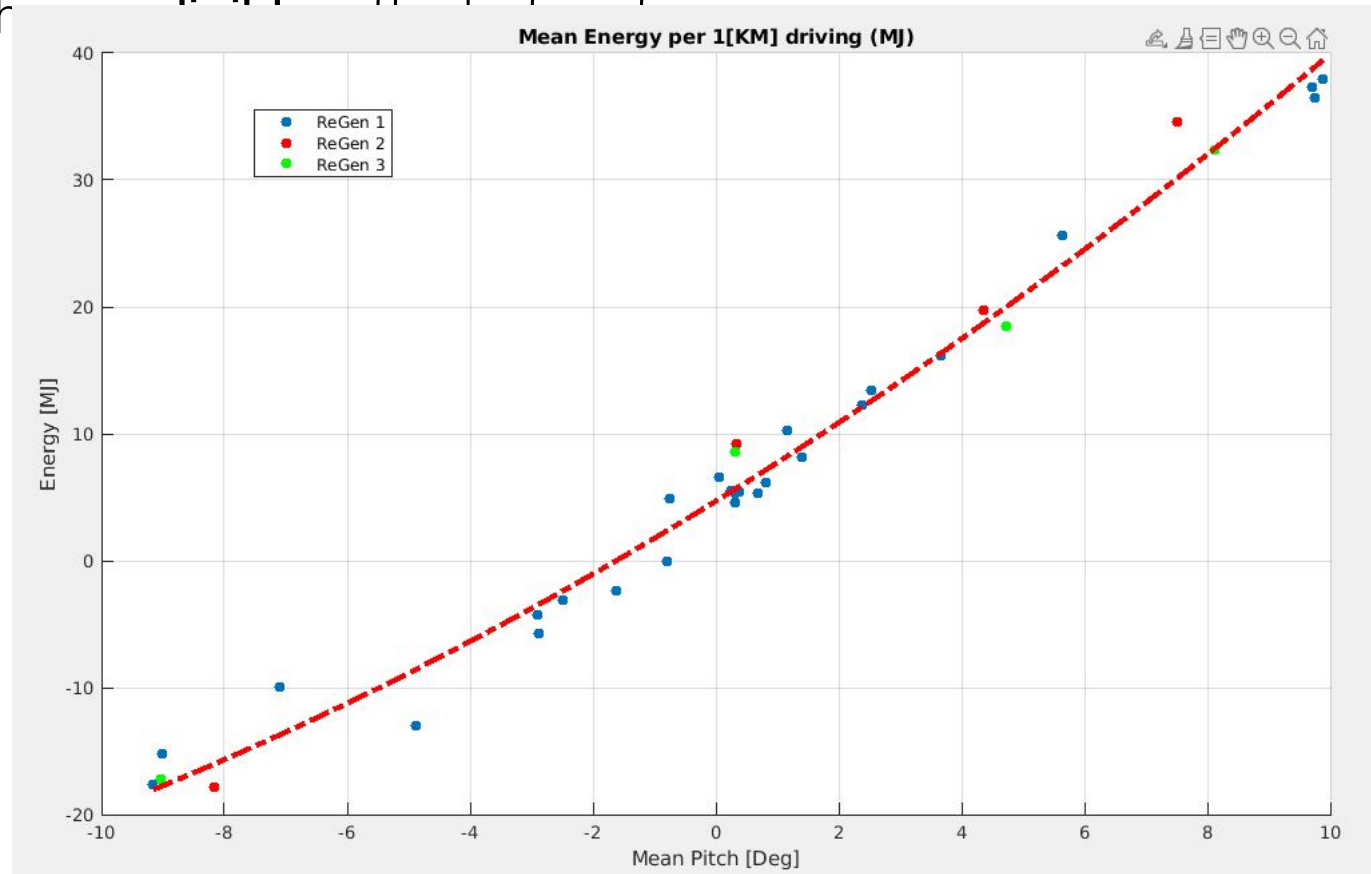
V. Midterm research results

Research results	Characteristics/Performance	Remark
Mid-term report	- Collected data on a Hyundai IONIQ 5 for many driving scenarios on different road grades - Analyzed the data and suggested a model that can fit the data - identify weak points of the model	
Final Report		
H/W		
S/W	- Implement a model that predicts the energy consumption based on road grade and other driving parameters - Develop analysis tool to generate a model to predict energy consumption of EV on sloped roads - Implement a path planner that can take into account energy consumption based on road grade	
Theses		☐ submission ☐ Published ☐ Expected date (00.00.00)
	HYUNDAI MOTOR GROUP – 000 University	☐ Application

IV. Research contents – Data Collection

Data was collected through different grade roads and different ReGen modes.

- The mean Energy it takes to drive 1[Km] at a specific road slope
- The measurements seem to fit a parabolic model
- Going uphill and down hill is not the same (not bias)
- GeReg mode h



IV. Research contents – Generalizing the energy prediction model

- Identifying the main parameters that influence the energy consumption of the car in sloped roads can be done from physical understanding and academic papers

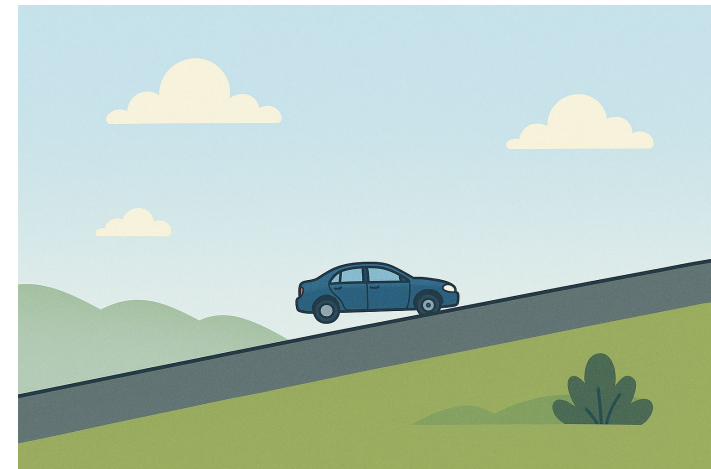
Basic model (for Ioniq 5):

$$E_{12} = (A_m \cdot g_{avg}^2 + B_m \cdot g_{avg} + C_m)L_{12}$$

Where

A_m, B_m, C_m are parameters of the model as extracted from test
 g_{avg} is the average grade along the path between 1 and 2
 L_{12} is the length of the path between 1 and 2

- Validating the energy prediction model will require data collection from other cars in different configuration to validate



IV. Research contents – Generalizing the energy prediction model

- Papers used in this work:
 - Electric vehicle's electricity consumption on a road with different slope (by Yang et al., 2014) – tries to adapt a theoretical model to empirical results
 - Electric vehicle energy consumption modelling and prediction based on road information (by J. Wang et al., 2015) – A validated five-part model predicts EV energy consumption using road, weather, and traffic
 - Strategies for the Modelisation of Electric Vehicle Energy Consumption (Di Martino et al., 2022) – It reviews and categorises the main modelling strategies for electric-vehicle energy consumption

IV. Research contents – Generalizing the energy prediction model

- The main parameters that can affect the energy consumption:
 - Mass of vehicle (M) – have linear effect on energy consumption
 - Engine Efficiency (e) – efficiency in positive slopes and negative slopes (ReGen)
 - Speed of Car
 - second order terms – temperature, road quality, altitude

General model (relative to Ioniq 5):

$$E_{12} = (A_m \cdot g_{avg}^2 + B_m \cdot g_{avg} + C_m) L_{12} \cdot \frac{M}{M_m} \cdot \left(\frac{\eta_e}{\eta_m}\right)^2 \cdot \left(1 + \alpha_V \frac{V - V_m}{V_m}\right)$$

Where

A_m, B_m, C_m are parameters pf the model as extracted from test

g_{avg} is the average grade along the path between 1 and 2

L_{12} is the length of the path between 1 and 2

η_e is energy efficiency

V Velocity (model velocity is about 25[mph])

α_V empirical parameter

VI. Future research plan

1. Next steps

- 1) Incorporate the energy model suggested in a path planner to see the effect
- 2) Try to validate the model by introducing changes to the car (add weight, drag?)
- 3) Validate the model with open source car simulations that may have slope-aware energy models

